Recall Task:
Predicting house prices

©2015 Emily Fox & Carlos Guestrin

Troy University Resources

How much is my house worth?



©2015 Emily Fox & Carlos Guestrin

How much is my house worth?



©2015 Emily Fox & Carlos Guestrin

Troy University Resources

Look at recent sales in my neighborhood

- How much did they sell for?



©2015 Emily Fox & Carlos Guestrin

Regression fundamentals: data, model, task

©2015 Emily Fox & Carlos Guestrin

Troy University Resources

Data

input *output*



$(x_1 = \text{sq.ft.}, y_1 = \$)$



$(x_2 = \text{sq.ft.}, y_2 = \$)$



$(x_3 = \text{sq.ft.}, y_3 = \$)$



$(x_4 = \text{sq.ft.}, y_4 = \$)$



$(x_5 = \text{sq.ft.}, y_5 = \$)$

⋮

©2015 Emily Fox & Carlos Guestrin

Data

input *output*



$(x_1 = \text{sq.ft.}, y_1 = \$)$



$(x_2 = \text{sq.ft.}, y_2 = \$)$



$(x_3 = \text{sq.ft.}, y_3 = \$)$



$(x_4 = \text{sq.ft.}, y_4 = \$)$



$(x_5 = \text{sq.ft.}, y_5 = \$)$

⋮

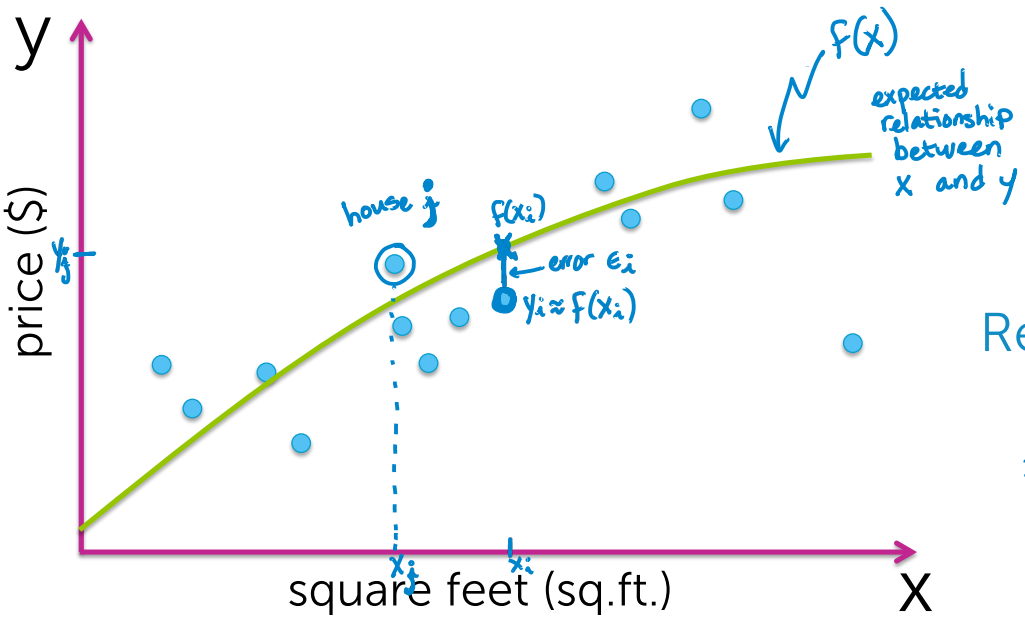
Input vs. Output:

- **y** is the quantity of interest
- assume **y** can be predicted from **x**

©2015 Emily Fox & Carlos Guestrin

Troy University Resources

Model –
How we *assume* the world works



Regression model:

$y_i = f(x_i) + \epsilon_i$

$E[\epsilon_i] = 0$ ← equally likely that error is + or −

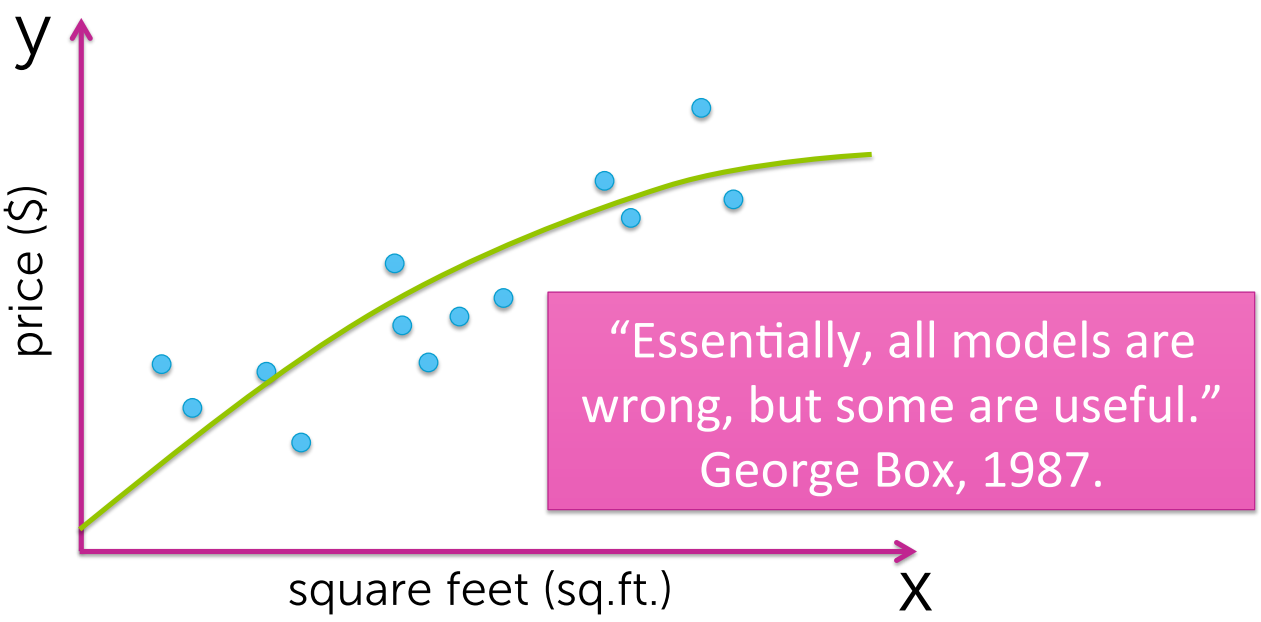
↕ expected value

⇓

y_i is equally likely to be above or below $f(x_i)$

©2015 Emily Fox & Carlos Guestrin

Model –
How we *assume* the world works

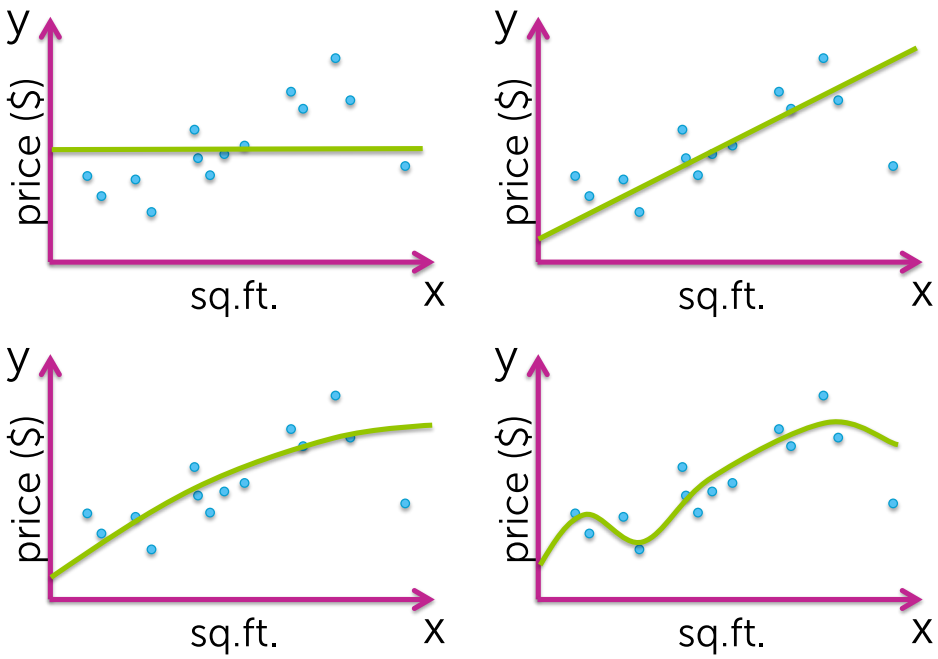


"Essentially, all models are wrong, but some are useful."
George Box, 1987.

©2015 Emily Fox & Carlos Guestrin

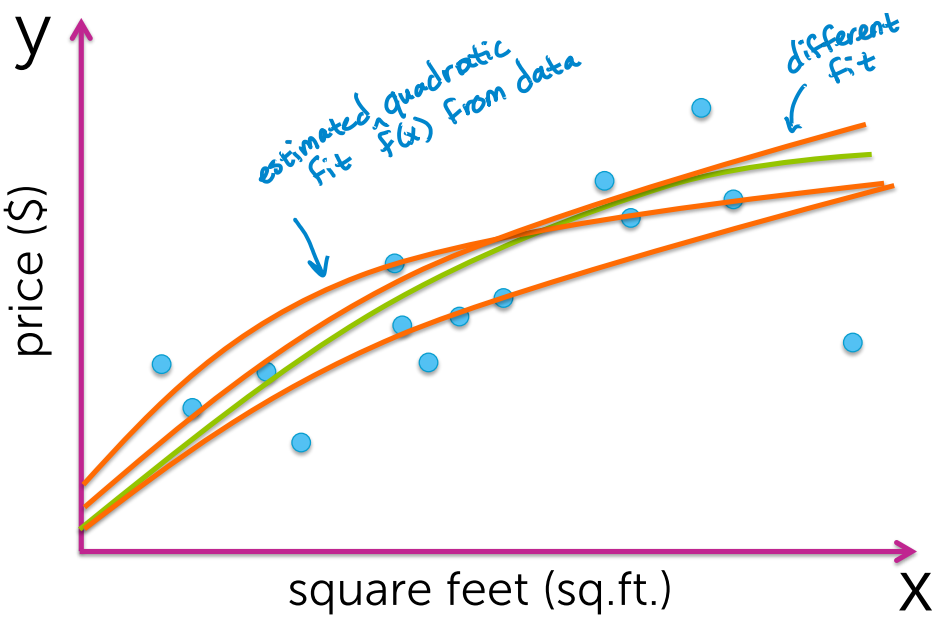
Troy University Resources

Task 1–
Which model f(x)?



©2015 Emily Fox & Carlos Guestrin

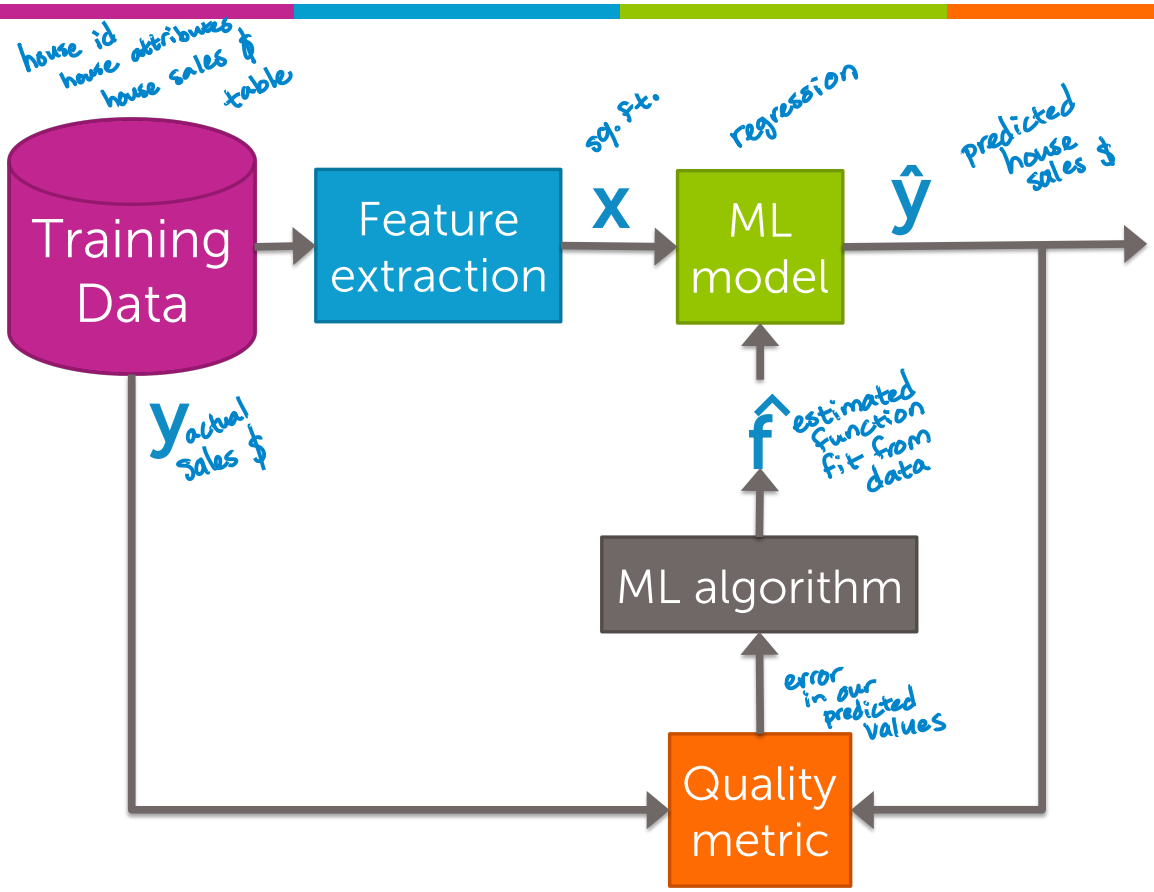
Task 2 – For a given model f(x), estimate function $\hat{f}(x)$ from data



Assume model f(x) is a quadratic function

©2015 Emily Fox & Carlos Guestrin

Troy University Resources

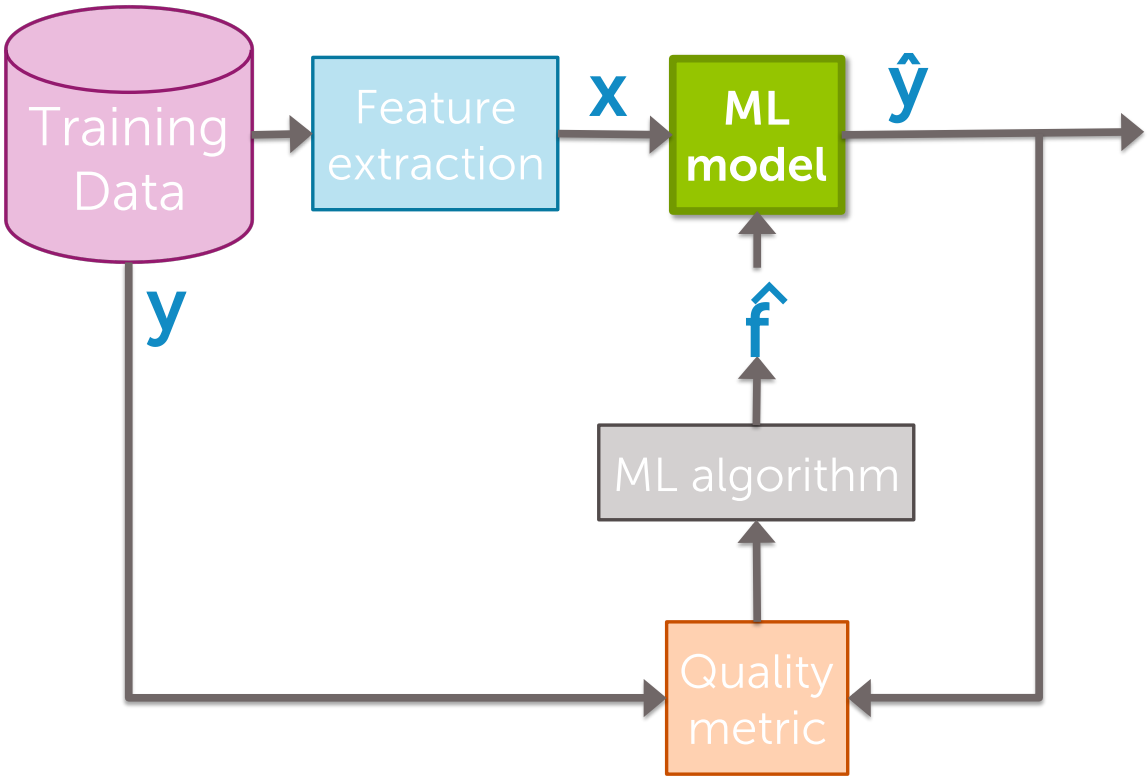


©2015 Emily Fox & Carlos Guestrin

Simple linear regression

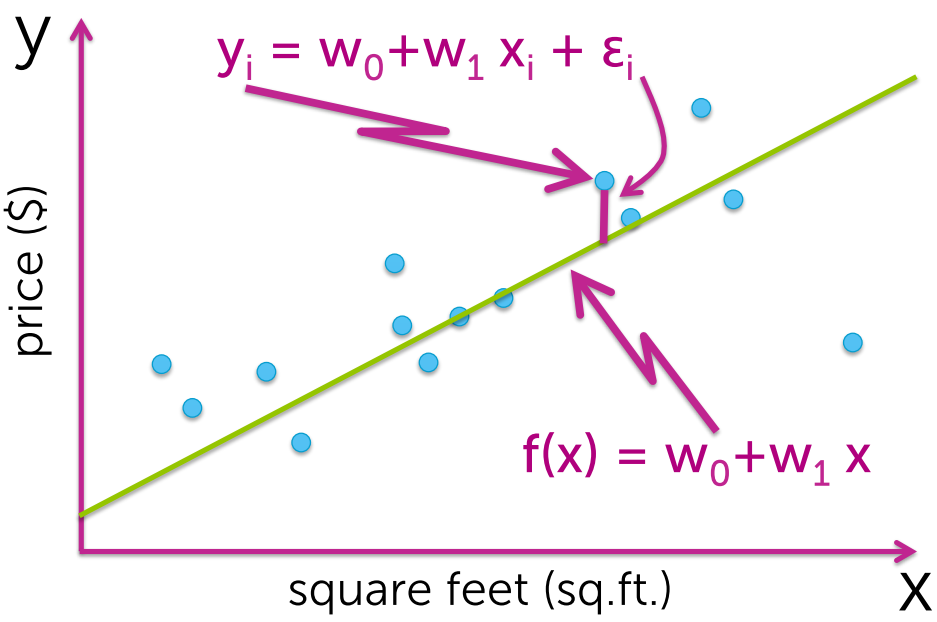
©2015 Emily Fox & Carlos Guestrin

Troy University Resources



©2015 Emily Fox & Carlos Guestrin

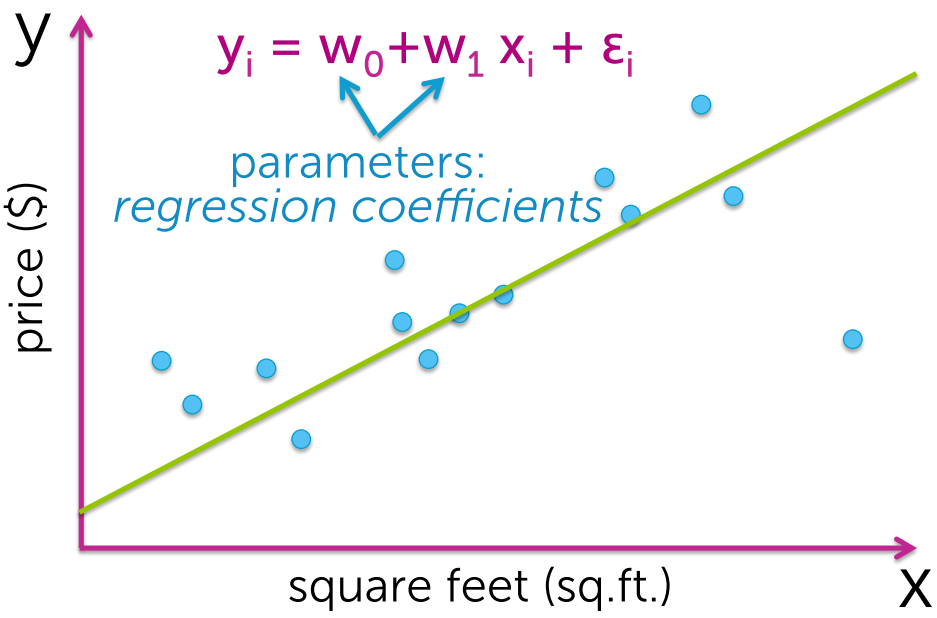
Simple linear regression model



©2015 Emily Fox & Carlos Guestrin

Troy University Resources

Simple linear regression model

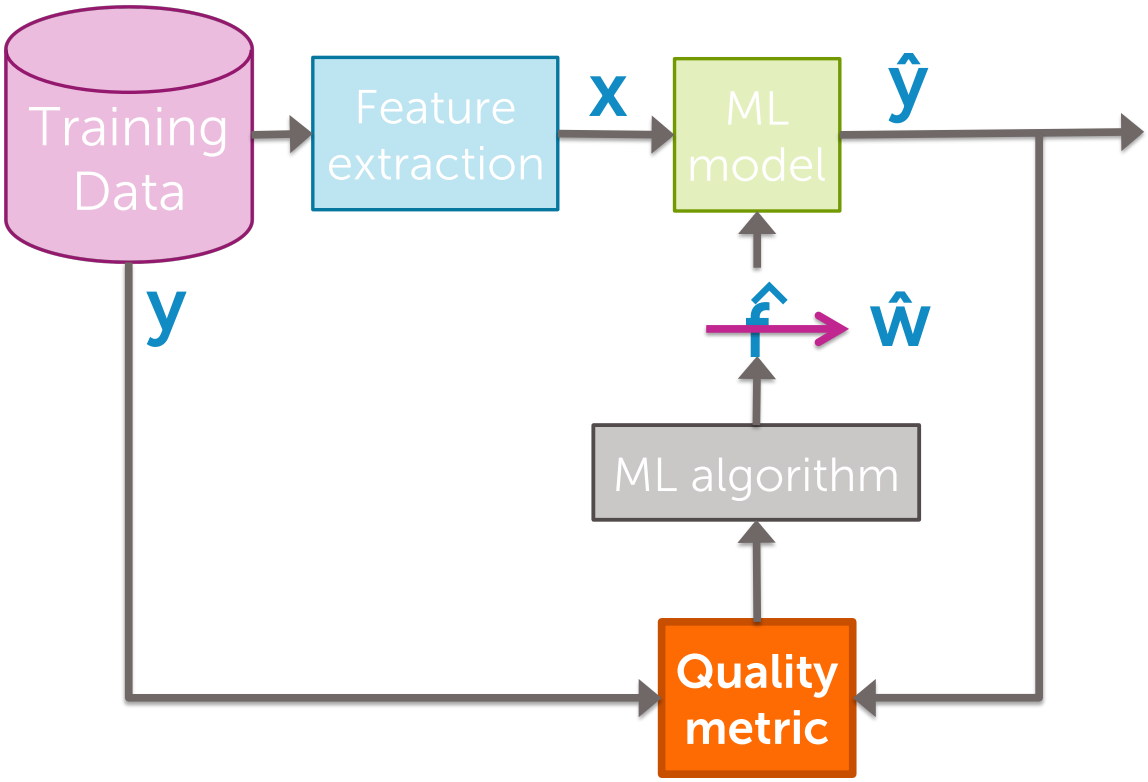


©2015 Emily Fox & Carlos Guestrin

Fitting a line to data

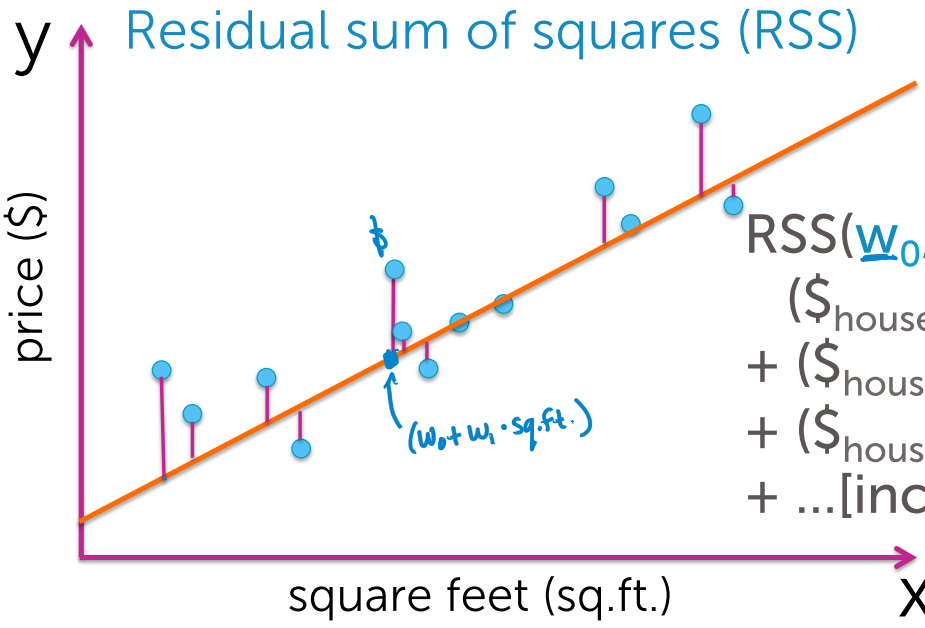
©2015 Emily Fox & Carlos Guestrin

Troy University Resources



©2015 Emily Fox & Carlos Guestrin

"Cost" of using a given line

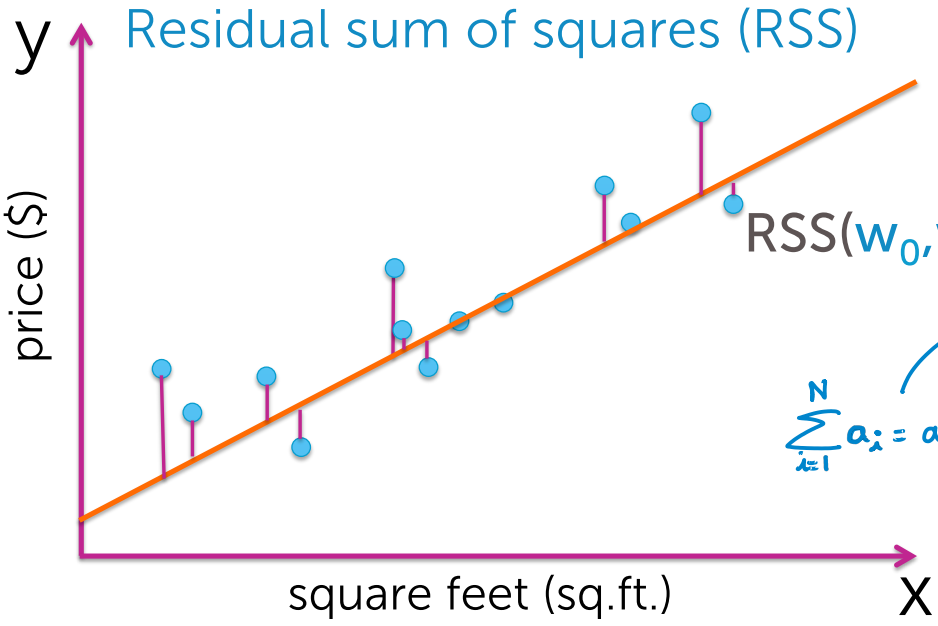


$$RSS(w_0, w_1) = (\$_{house\ 1} - [w_0 + w_1 sq.ft._{house\ 1}])^2 + (\$_{house\ 2} - [w_0 + w_1 sq.ft._{house\ 2}])^2 + (\$_{house\ 3} - [w_0 + w_1 sq.ft._{house\ 3}])^2 + \ldots [include\ all\ training\ houses]$$

©2015 Emily Fox & Carlos Guestrin

Troy University Resources

"Cost" of using a given line



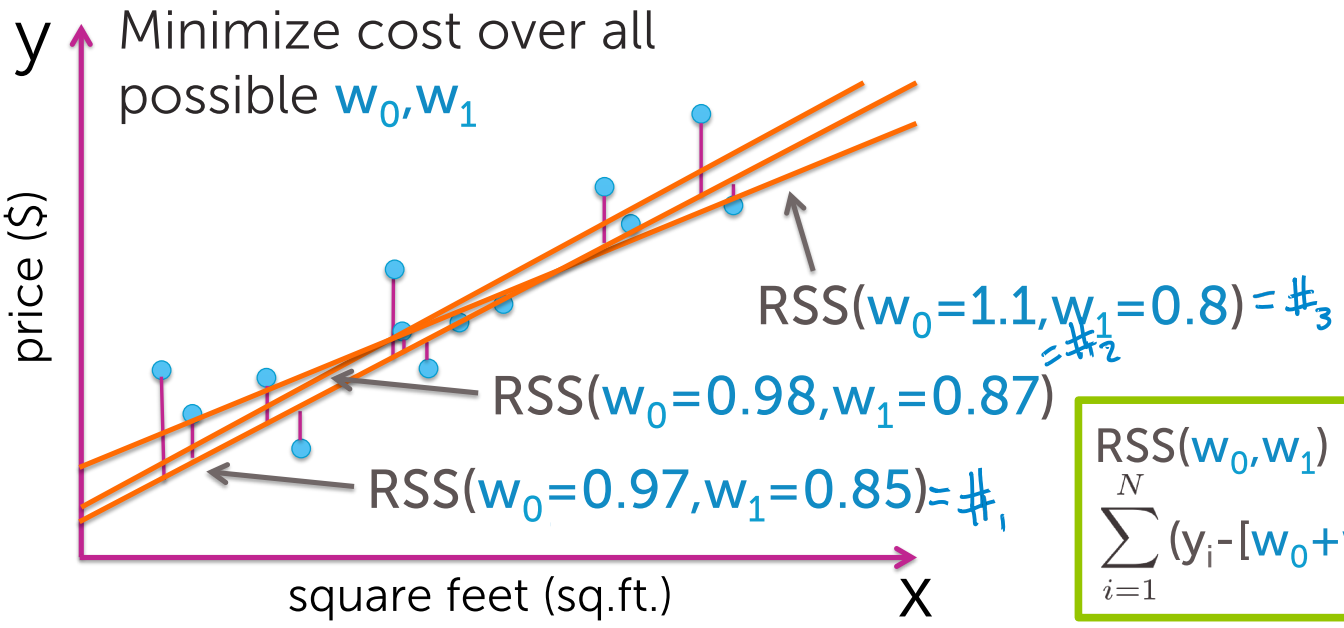
$$RSS(w_0, w_1) = \sum_{i=1}^{N} (y_i - [w_0 + w_1 x_i])^2$$

$\sum_{i=1}^{N} a_i = a_1 + a_2 + \ldots + a_N$

Here, $a_i = (y_i - [w_0 + w_1 x_i])^2$

©2015 Emily Fox & Carlos Guestrin

Find "best" line



$$RSS(w_0, w_1) = \sum_{i=1}^{N} (y_i - [w_0 + w_1 x_i])^2$$

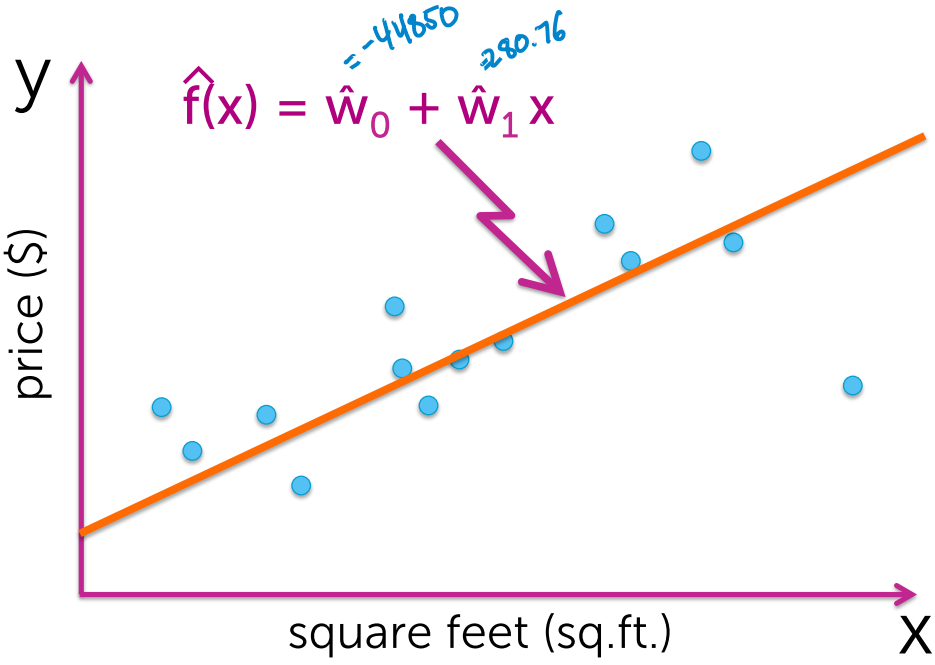
©2015 Emily Fox & Carlos Guestrin

Troy University Resources

The fitted line: use + interpretation

©2015 Emily Fox & Carlos Guestrin

Model vs. fitted line



Regression model:

$$y_i = w_0 + w_1 x_i + \varepsilon_i$$

parameters (unknown variables)

Estimated parameters:

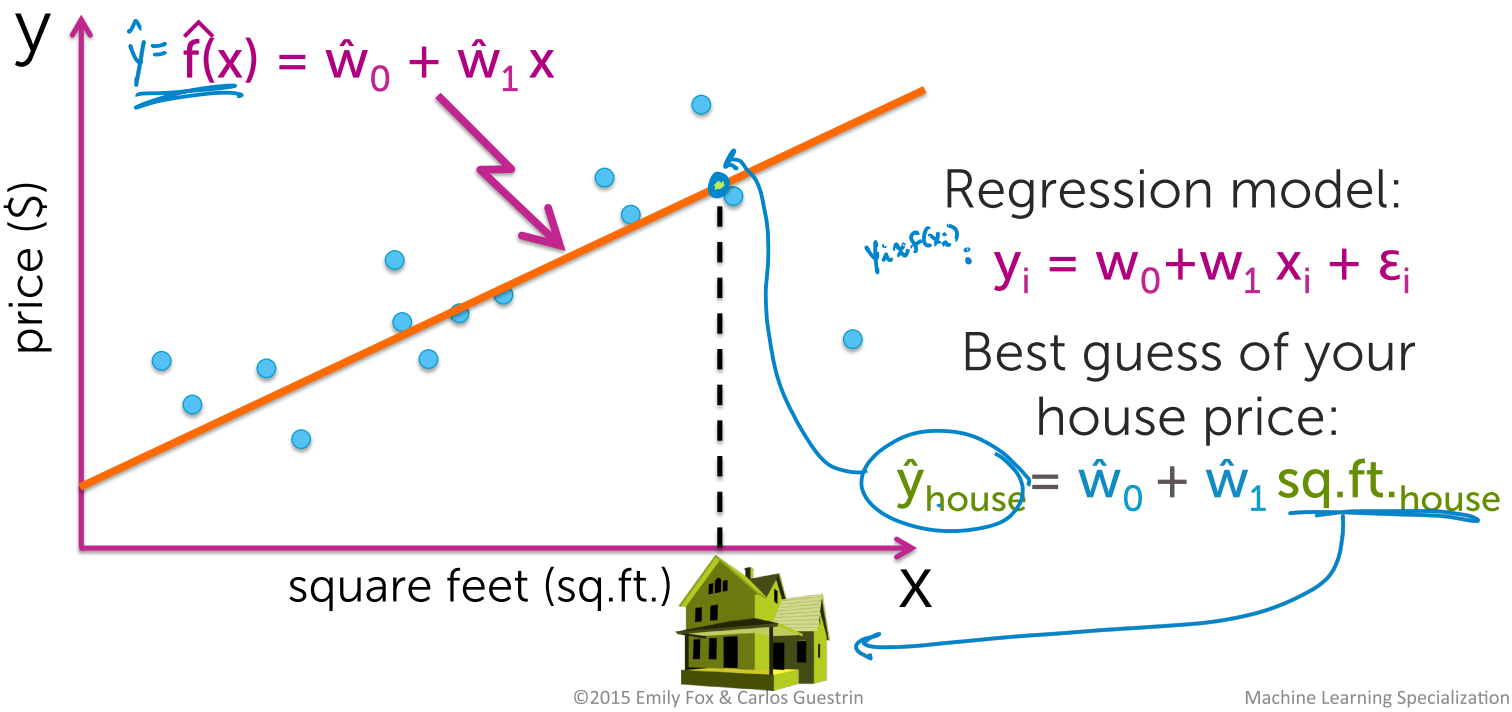
$\hat{w}_0 \approx -44850$, $\hat{w}_1 \approx 280.76$

take actual values

©2015 Emily Fox & Carlos Guestrin

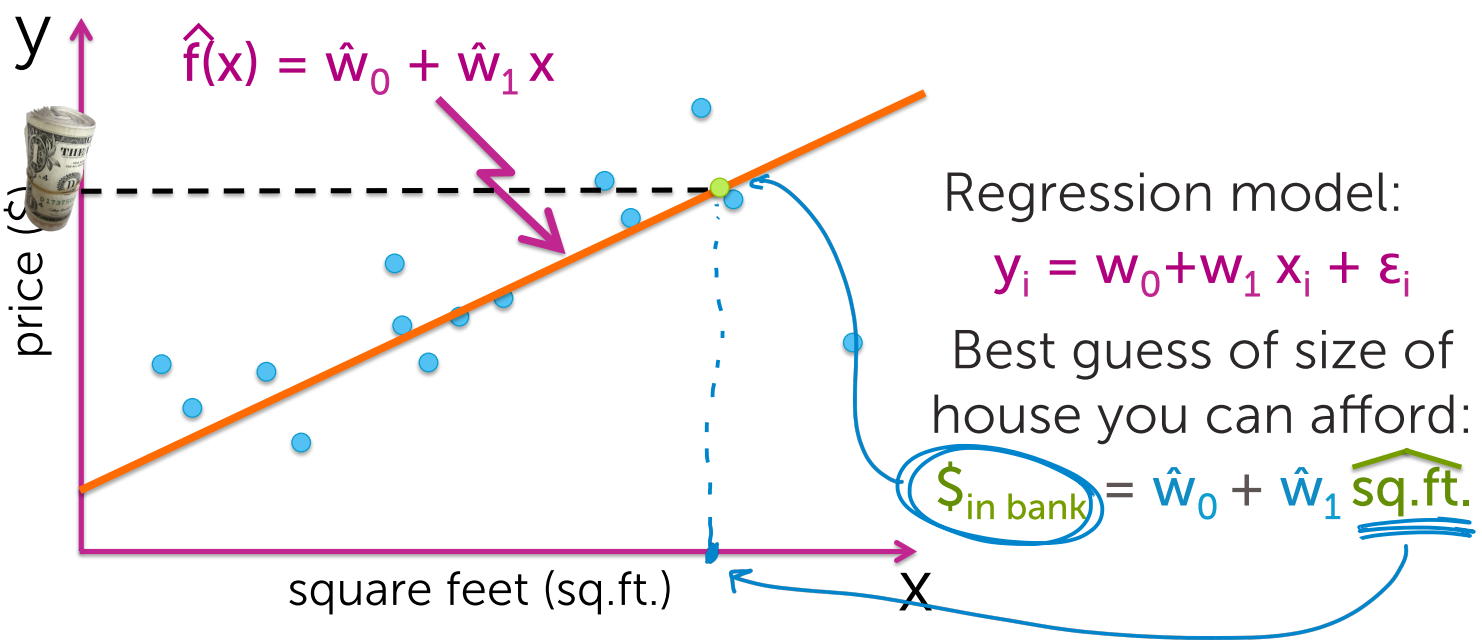
Troy University Resources

Seller:
Predicting your house price



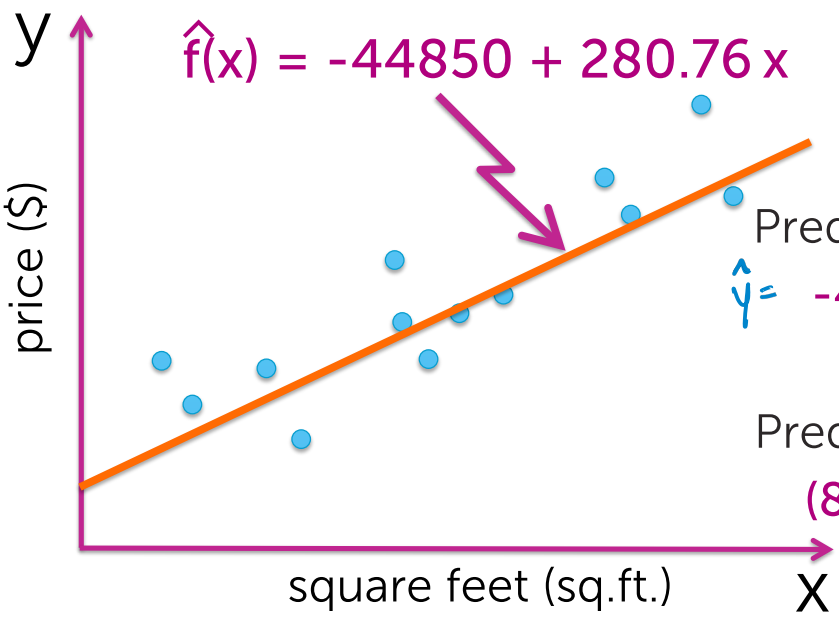
©2015 Emily Fox & Carlos Guestrin

Buyer:
Predicting size of house



©2015 Emily Fox & Carlos Guestrin

A concrete example



Predict $ of 2,640 sq.ft. house:

$\hat{y}$ = -44850 + 280.76 * 2,640

= $696,356

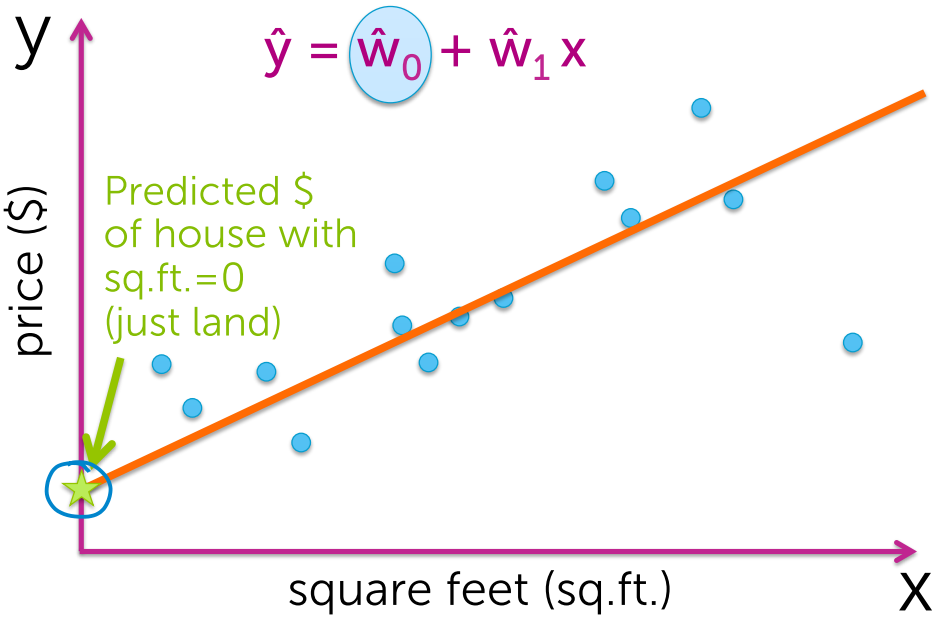
Predict sq.ft. of $859,000 sale:

(859000+44850)/ 280.76

= 3,219 sq.ft.

©2015 Emily Fox & Carlos Guestrin

Interpreting the coefficients



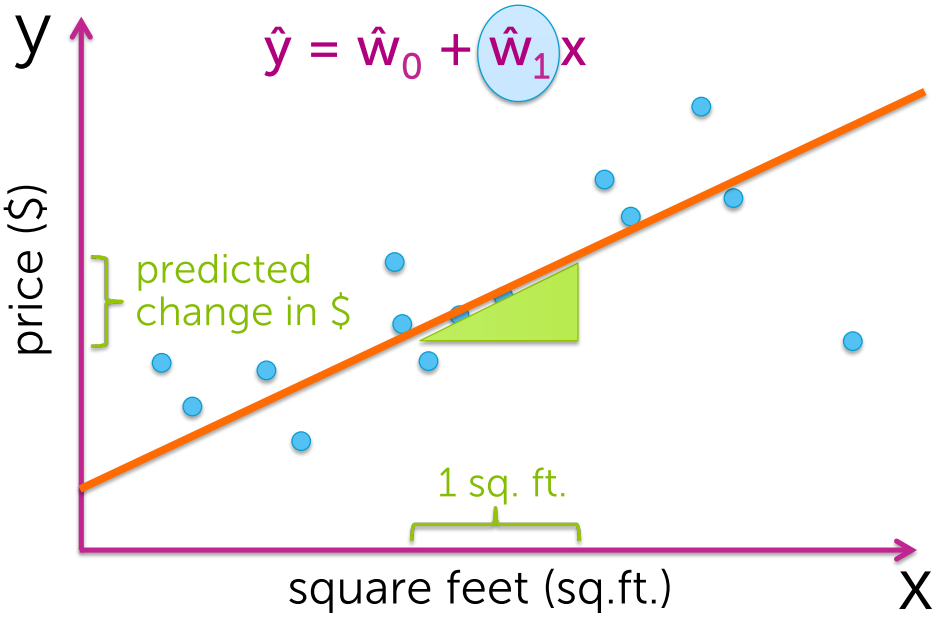
$\hat{y} = \hat{w}_0$ when $x = 0$

not very meaningful

©2015 Emily Fox & Carlos Guestrin

Troy University Resources

Interpreting the coefficients



$\hat{\$}_{1001 \text{ sq.ft.}} - \hat{\$}_{1000 \text{ sq.ft.}}$

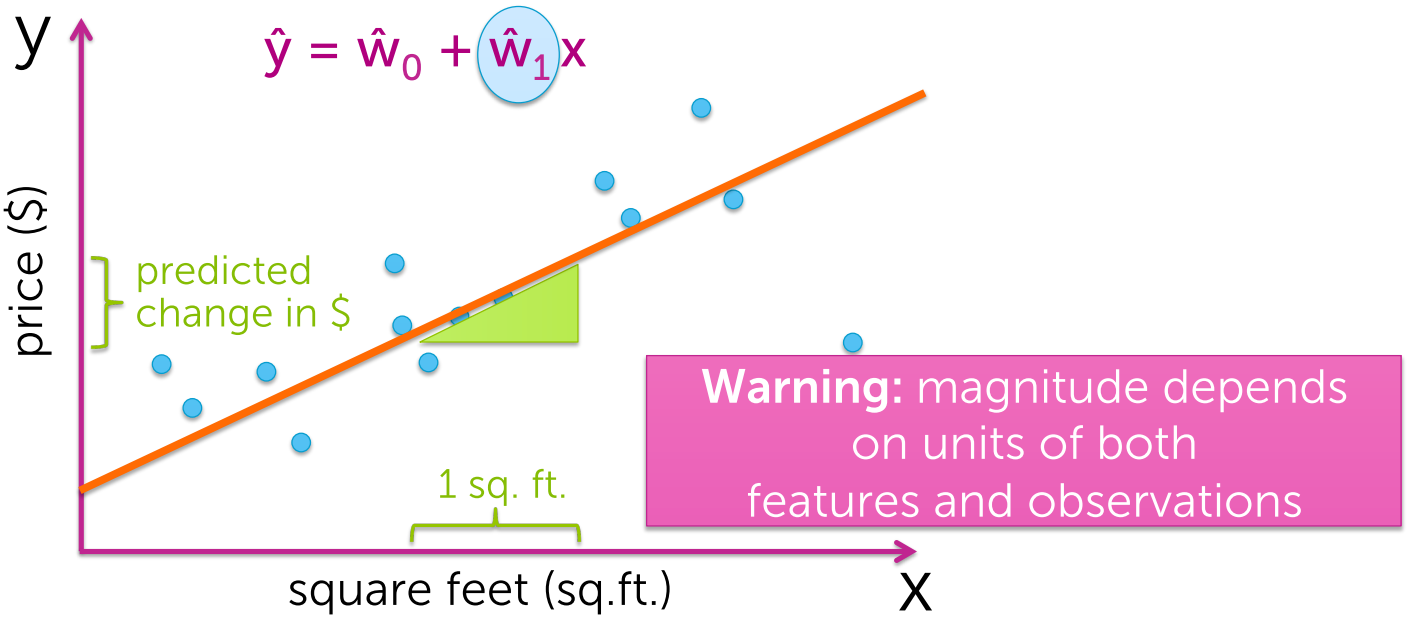
$= \hat{w}_0 + \hat{w}_1 \cdot 1001 \text{ sq.ft.} - (\hat{w}_0 + \hat{w}_1 1000 \text{ sq.ft.})$

$= \hat{w}_1$

predicted change in the output per unit change in input

©2015 Emily Fox & Carlos Guestrin

Interpreting the coefficients

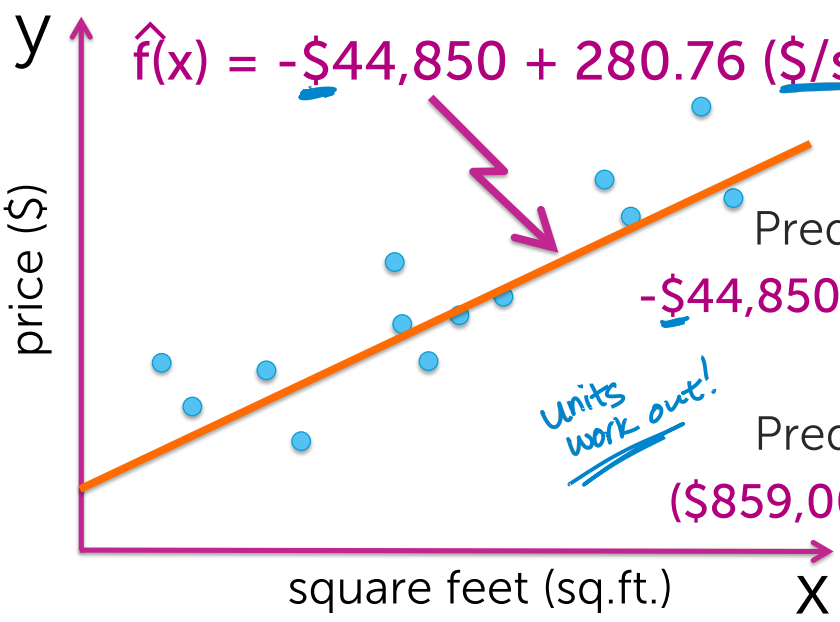


Warning: magnitude depends on units of both features and observations

©2015 Emily Fox & Carlos Guestrin

Troy University Resources

A concrete example



$\hat{f}(x)$ = -$44,850 + 280.76 ($/sq.ft.) x

Predict $ of 2,640 sq.ft. house:

-$44,850 + 280.76 ($/sq.ft.) * 2,640 sq.ft.

= $696,356

Units work out!

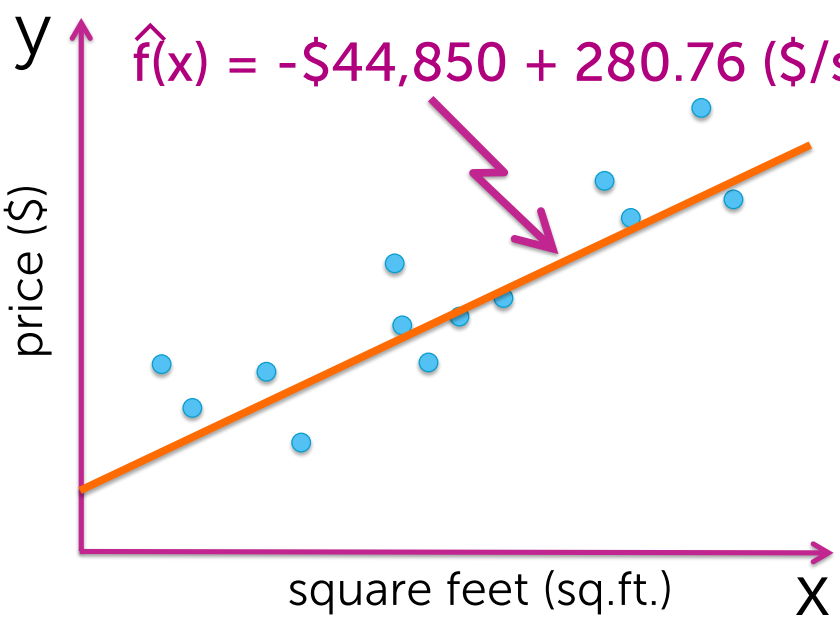
Predict sq.ft. of $859,000 sale:

($859,000+$44,850)/ 280.76 ($/sq.ft.)

= 3,219 sq.ft.

©2015 Emily Fox & Carlos Guestrin

A concrete example



But what if:

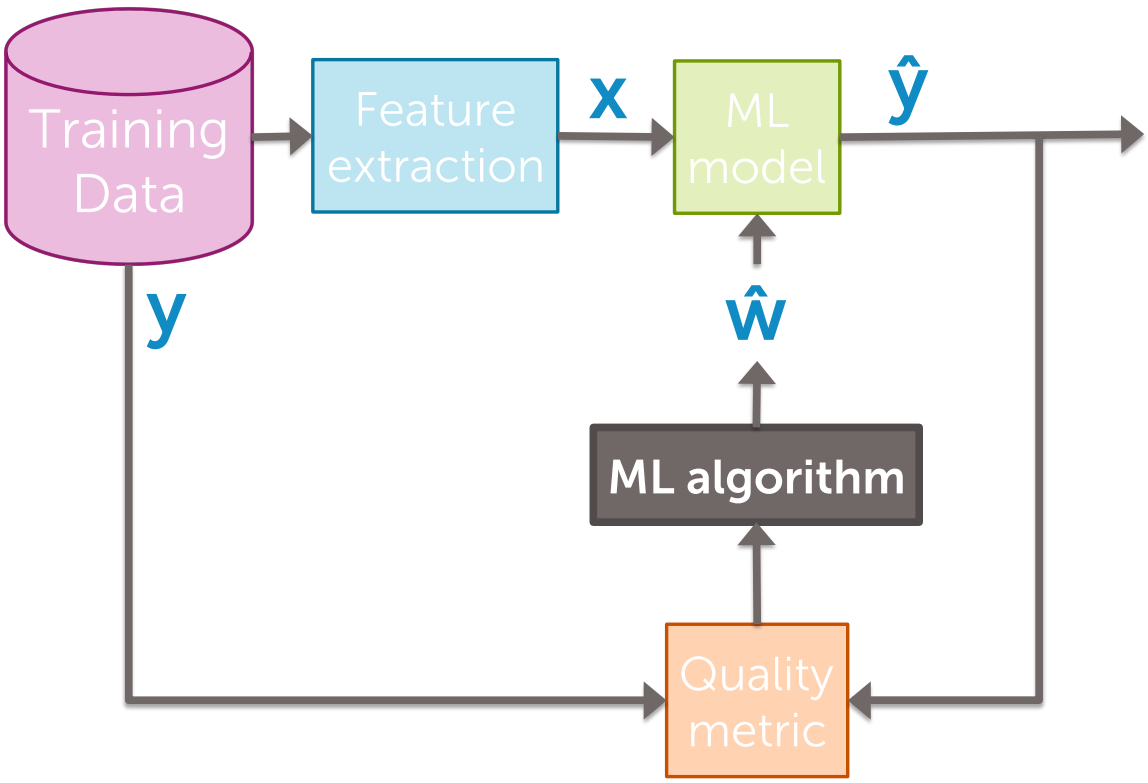
- House was measured in square meters?
- Price was measured in RMB?

©2015 Emily Fox & Carlos Guestrin

Troy University Resources

Algorithms for fitting the model

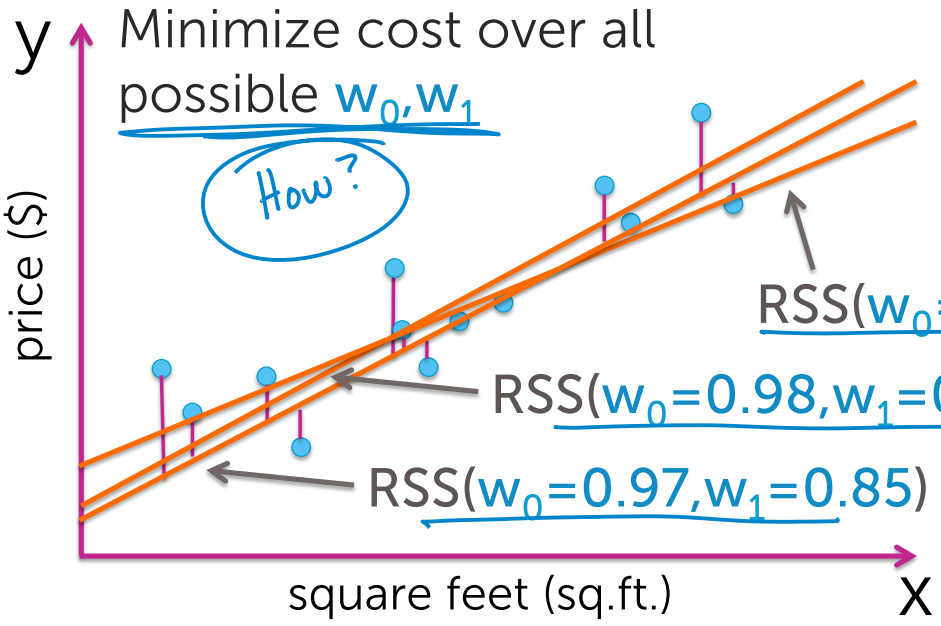
©2015 Emily Fox & Carlos Guestrin



©2015 Emily Fox & Carlos Guestrin

Troy University Resources

Find "best" line

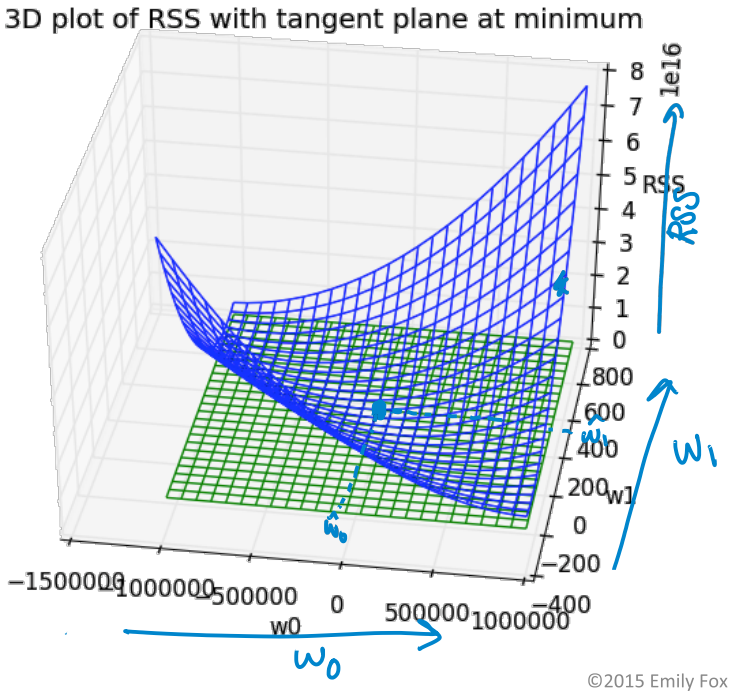


Recall:

$$RSS(w_0, w_1) = \sum_{i=1}^{N} (y_i - [w_0 + w_1 x_i])^2$$

©2015 Emily Fox & Carlos Guestrin

Minimizing the cost



Minimize function over all possible w_0, w_1

$$\min_{w_0,w_1} \sum_{i=1}^{N} (y_i - [w_0 + w_1 x_i])^2$$

RSS(w_0,w_1) is a function of 2 variables = $g(w_0, w_1)$

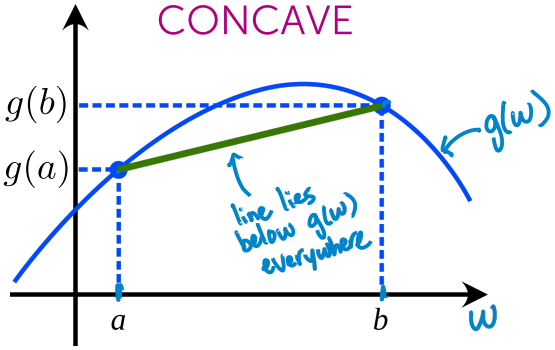
©2015 Emily Fox & Carlos Guestrin

An aside on optimization

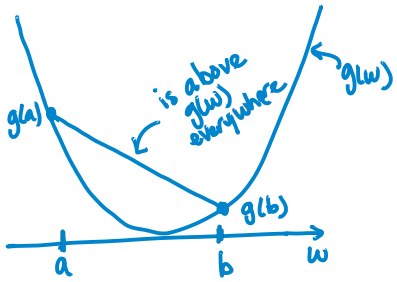
©2015 Emily Fox & Carlos Guestrin

Convex/concave functions

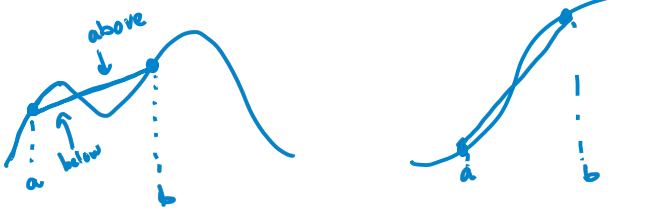
CONCAVE



CONVEX



NEITHER

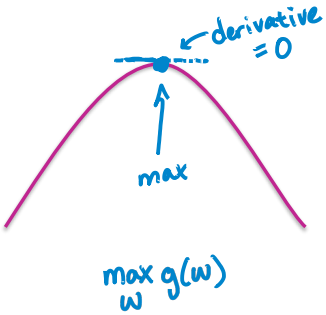


©2015 Emily Fox & Carlos Guestrin

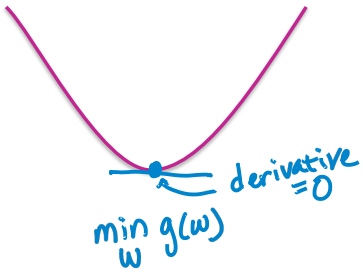
Troy University Resources

Finding the max or min analytically

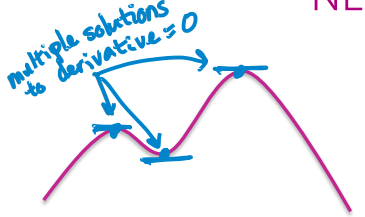
CONCAVE

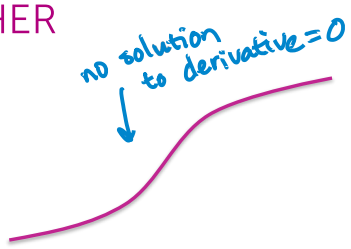


CONVEX



NEITHER





Example:

$g(w) = 5-(w-10)^2$

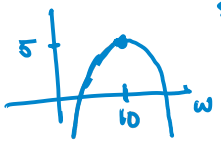
$\frac{dg(w)}{dw} = 0 - 2(w-10)^1 \cdot 1$

$= -2w + 20$

Set derivative = 0:

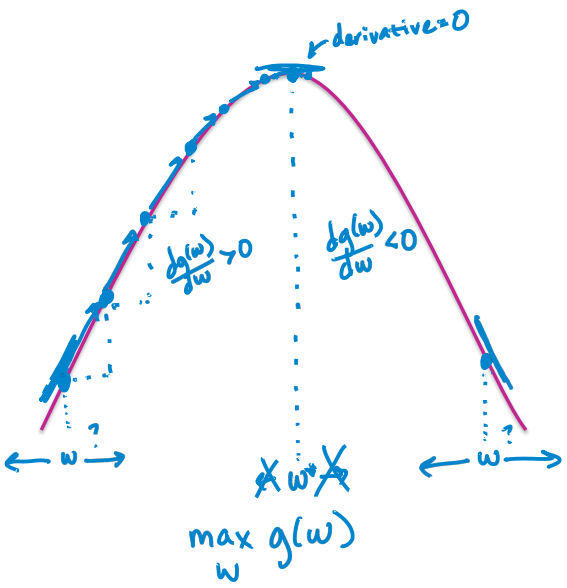
$-2w + 20 = 0$

$w = 10$



©2015 Emily Fox & Carlos Guestrin

Finding the max via hill climbing



How do we know whether to move w to right or left? (Inc. or dec. the value of w?)

while not converged

$$w^{(t+1)} \leftarrow w^{(t)} + \eta \frac{dg(w)}{dw}$$

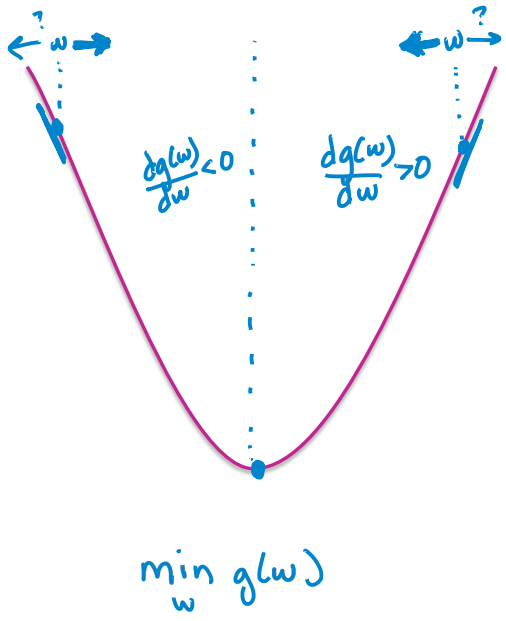
iteration t

stepsize

©2015 Emily Fox & Carlos Guestrin

Troy University Resources

Finding the min via hill descent



When derivative is positive, we want to decrease w and when derivative is negative, we want to increase w

Algorithm:

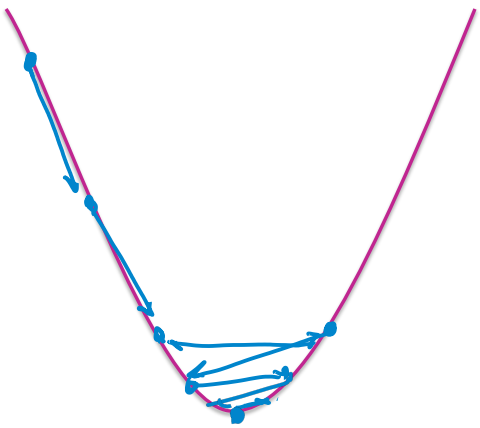
while not converged

$$w^{(t+1)} \leftarrow w^{(t)} - \eta \left.\frac{dg}{dw}\right|_{w^{(t)}}$$

©2015 Emily Fox & Carlos Guestrin

Choosing the stepsize—
Fixed stepsize

η



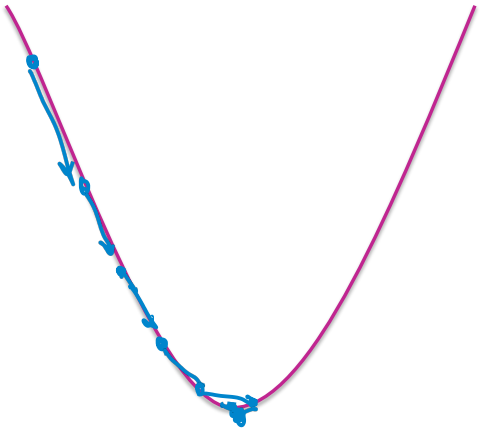
$\eta = 0.1$

©2015 Emily Fox & Carlos Guestrin

Troy University Resources

Choosing the stepsize—
Decreasing stepsize

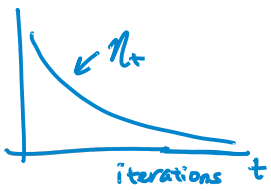
or stepsize schedule



Common choices:

$$\eta_t = \frac{\alpha}{t}$$

$$\eta_t = \frac{\alpha}{\sqrt{t}}$$



©2015 Emily Fox & Carlos Guestrin

Convergence criteria

For convex functions, optimum occurs when

$$\frac{dg(w)}{dw} = 0$$

In practice, stop when

$$\left|\frac{dg(w)}{dw}\right| < \epsilon$$

↑ threshold to be set

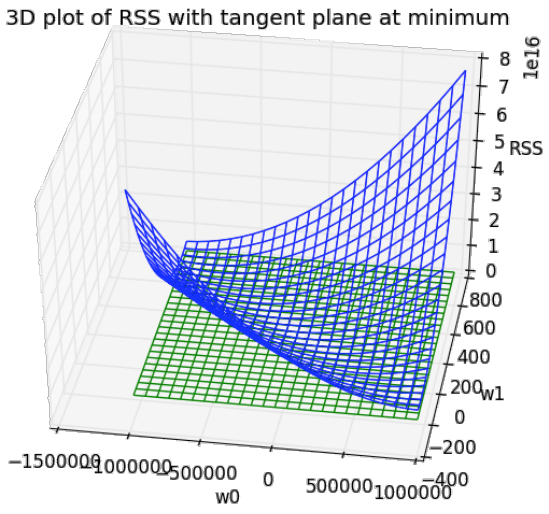
Algorithm:

while not converged

$$w^{(t+1)} \leftarrow w^{(t)} - \eta \left.\frac{dg}{dw}\right|_{w^{(t)}}$$

©2015 Emily Fox & Carlos Guestrin

Moving to multiple dimensions: Gradients



$$\nabla g(w) = \begin{bmatrix} \frac{\partial g}{\partial w_0} \\ \frac{\partial g}{\partial w_1} \\ \vdots \\ \frac{\partial g}{\partial w_p} \end{bmatrix}$$

gradient

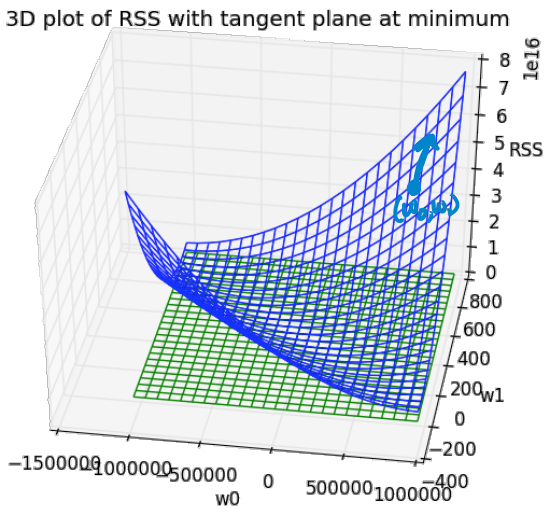
$[w_0, w_1, \ldots, w_p]$

$(p+1)$-dimensional vector

partial derivative is like a derivate with respect to w_1 treating all other variables as constants

©2015 Emily Fox & Carlos Guestrin

Gradient example



$$g(w) = 5w_0 + 10w_0w_1 + 2w_1^2$$

$$\frac{\partial g}{\partial w_0} = 5 + 10w_1$$

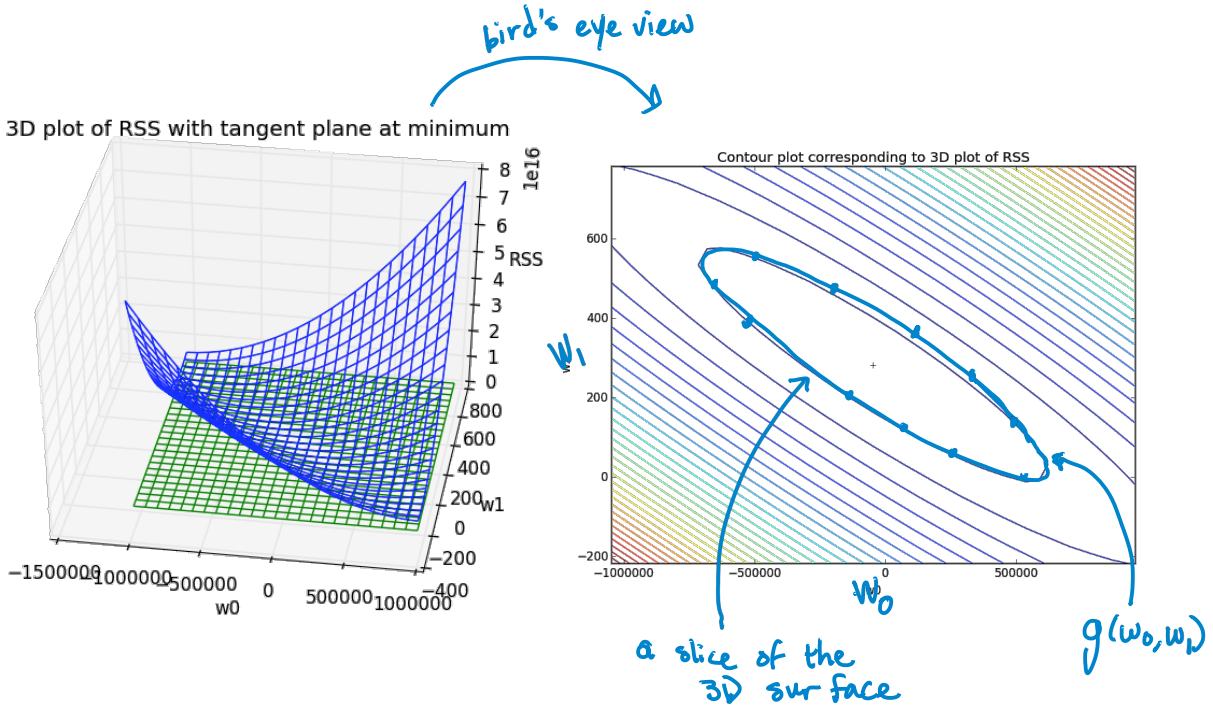
$$\frac{\partial g}{\partial w_1} = 10w_0 + 4w_1$$

$$\nabla g(w) = \begin{bmatrix} 5 + 10w_1 \\ 10w_0 + 4w_1 \end{bmatrix}$$

©2015 Emily Fox & Carlos Guestrin

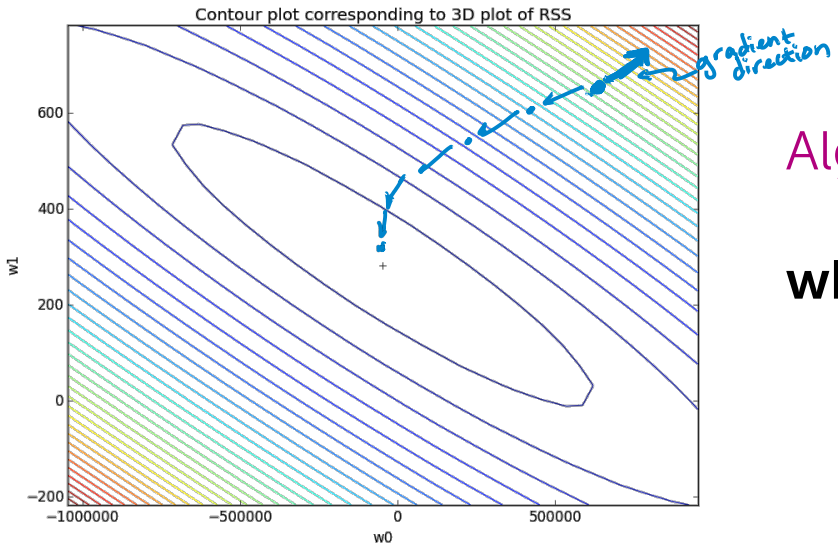
Troy University Resources

Contour plots



©2015 Emily Fox & Carlos Guestrin

Gradient descent



Algorithm:

while not converged

$$w^{(t+1)} \leftarrow w^{(t)} - \eta \nabla g(w^{(t)})$$

$$\begin{bmatrix} \\ \\ \end{bmatrix} \leftarrow \begin{bmatrix} \\ \\ \end{bmatrix} - \eta \begin{bmatrix} \\ \\ \end{bmatrix}$$

Convergence: $\|\nabla g(w)\| < \epsilon$

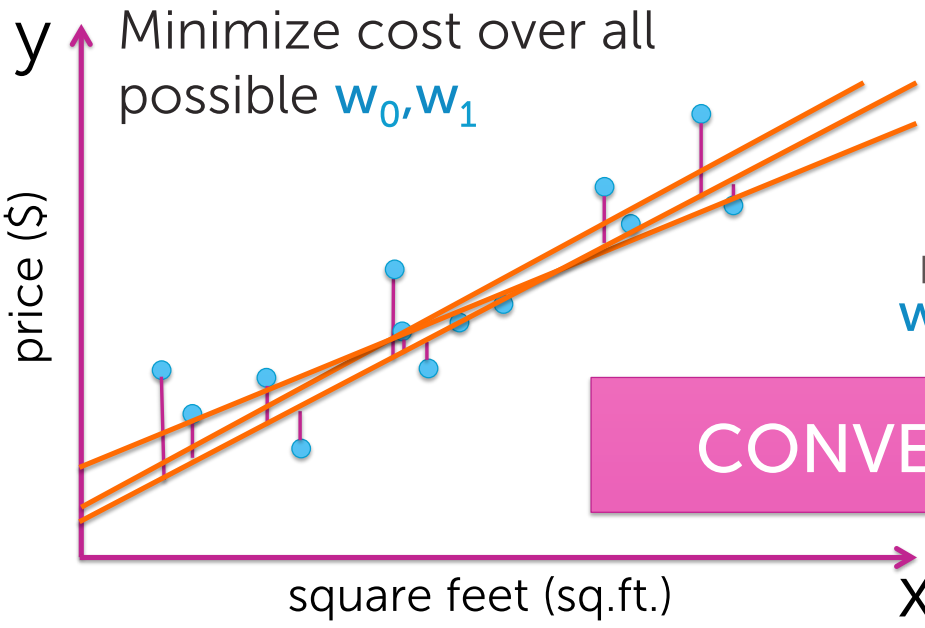
©2015 Emily Fox & Carlos Guestrin

Troy University Resources

Finding the least squares line

©2015 Emily Fox & Carlos Guestrin

Find "best" line



$$\min_{w_0, w_1} \sum_{i=1}^{N} (y_i - [w_0 + w_1 x_i])^2$$

CONVEX

⇒ solution is unique

+ gradient descent alg. will converge to minimum

©2015 Emily Fox & Carlos Guestrin

Troy University Resources